香港中文大學（深圳）
The Chinese University of Hong Kong, Shenzhen

Introduction to Data Science

Lecture 13: Midterm Review (Continued)

Outline

- Midterm Review: Practice Questions
- Midterm Review: Core Concept

Midterm Review: Practice Questions

Practice Question II:

4. **(25 points)** Poisson Distribution is a robust probability model that has widely used in Queue Theory. Suppose in the six adjacent and independent time period, the number of phone calls center got follows Poisson distribution, which are 22, 17, 18, 45, 6, 33. Calculate the maximum likelihood estimate for the parameter λ in any identically independent time period. Note: the parameter λ shall be positive. If the parameter is λ , the probability mass function of a Poisson random variable $X = k$ is $\lambda^k \exp(-\lambda)/k!$.

8. The number of threes made during an NBA game is Poisson distribution. Last Saturday, the number of threes made were 14, 26, 25, and 13. Calculate the maximum likelihood estimate for the parameter λ. Note: the parameter λ shall be positive. If the parameter is λ , the probability mass function of a Poisson random variable $X = k$ is $\lambda^k \exp(-\lambda)/k!$.

- The first question is from our Quiz 1. The second question is from our assignment 2.

- What are these two questions asking for?
  - These two questions ask for the **Maximum Likelihood Estimate (MLE) of the parameter λ** for a Poisson distribution. Specifically, given a set of observed data points that are assumed to follow a Poisson distribution, we are asked to estimate the value of λ that maximizes the likelihood function.

- *Although the Poisson distribution was not explicitly covered in class, this does not prevent solving the problem*.
- Why?
  - Since the probability mass function (PMF) is given, and the question asks for the MLE, we can follow the standard MLE derivation to find the solution.

- Let’s solve these two questions together.
- 1. Given the question description, we know the Poisson PMF is:

$$P(X = k) = \frac{\lambda^k e^{-\lambda}}{k!}, \quad k = 0, 1, 2, \ldots$$

where λ in this question is the unknown model parameter. We are asked to estimate λ given observations using MLE.

2. Given n independent Poisson-distributed observations $X_1, X_2, \dots, X_n$

- in the first question, we have observed $X_1, X_2, \dots, X_n$ data recorded as: 22,17,18,45,6,33
- in the second question, we have observed $X_1, X_2, \dots, X_n$ data recorded as: 14,26,25, 13

MLE for Poisson: The key underlying logic is that, given n independent Poisson-distributed observations $X_1, X_2, \ldots, X_n$, the likelihood function is:

$$L(\lambda) = \prod_{i=1}^{n} \frac{\lambda^{X_i} e^{-\lambda}}{X_i!}$$

Taking the log-likelihood function:

$$\log L(\lambda) = \sum_{i=1}^{n} X_i \log \lambda - n\lambda + C$$

where C is a constant that does not depend on λ. Differentiating with respect to λ and setting to zero:

$$\frac{d}{d\lambda}\left(\sum_{i=1}^{n} X_i \log \lambda - n\lambda\right) = \frac{\sum X_i}{\lambda} - n = 0$$

Solving for λ,

$$\hat{\lambda} = \frac{1}{n}\sum_{i=1}^{n} X_i$$

That is, the **MLE for λ is simply the sample mean.**

For the first question:

- Given the data: $22, 17, 18, 45, 6, 33$
- Compute the sample mean:
$$\hat{\lambda} = \frac{22 + 17 + 18 + 45 + 6 + 33}{6}$$

For the second question:

- Given the data: $14, 26, 25, 13$
- Compute the sample mean:
$$\hat{\lambda} = \frac{14 + 26 + 25 + 13}{4}$$

Conclusion:

- Although the Poisson distribution was not explicitly covered in class, this does not prevent solving the problem. Since the probability mass function (PMF) is given, and the question asks for the MLE, we can follow the standard MLE derivation to find the solution.

- Suppose I continue to ask you to check the given

$$P(X = k) = \frac{\lambda^k e^{-\lambda}}{k!}, \quad k = 0, 1, 2, \ldots$$

is a valid PMF and ask you to compute the mean

$$E[X] = \sum_{k=0}^{\infty} kP(X = k).$$

How to solve the problems?

For example, to verify the validity of the PMF:

A function is a valid PMF if the sum of probabilities over all possible values equals 1:

$$\sum_{k=0}^{\infty} P(X=k) = \sum_{k=0}^{\infty} \frac{\lambda^k e^{-\lambda}}{k!} = 1.$$

This sum is the Taylor series expansion of e^{λ}:

$$\sum_{k=0}^{\infty} \frac{\lambda^k}{k!} = e^{\lambda}.$$

Multiplying by $e^{-\lambda}$:

$$e^{-\lambda} \sum_{k=0}^{\infty} \frac{\lambda^k}{k!} = e^{-\lambda} e^{\lambda} = 1.$$

Since the sum equals 1, the Poisson PMF is valid.

- As for computing $E(X)$ using

$$E[X] = \sum_{k=0}^{\infty} kP(X = k).$$

please derive by yourself….

Practice Question III:

(f) **(6 points)** Suppose the weights of randomly selected American female college students are normally distributed with unknown mean μ and standard deviation σ. A random sample of 6 American female college students yielded the following weights (in pounds): 115, 117, 128, 125, 151, 162. Suppose we also know the sampled students' ages (in the same order): 18, 19, 20, 20, 21, 22. Based on this additional information, use a linear regression to find the relationship between American female college students' weight and their age.

Exercise 3

You have data on the number of hours students studied and their corresponding scores on a test. Use linear regression to model the relationship between hours studied (independent variable) and test scores (dependent variable).

Student	1	2	3	4	5	6	7	8	9	10
Hours Studied	5.4	3.9	2.8	4.3	2.1	4.3	4.5	2.8	2.2	2.3
Test Scores	92.9	86.0	78.8	88.2	76.2	87.6	89.0	78.9	76.0	77.0

① Calculate the slope and intercept of the best-fit line using MLE.

② Use the linear model obtained in the previous exercise, predict the test score for a student who studied for 1 hours.

③ Perform a residual analysis on the linear regression model. Calculate the mean squared error, and plot the residuals versus the predicted values to check for any patterns that might indicate violations of linear regression assumptions.

- The first question is from last year's midterm. The second question is from our tutorial. To solve these questions, you will be given the following information.

- If $Y \sim \mathcal{N}(\beta_0 + \beta_1 X, \sigma^2)$, the MLE of β_0 and β_1 is as follows:

$$\hat{\beta}_1 = \frac{\sum_i (x_i - \bar{x})(y_i - \bar{y})}{\sum_i (x_i - \bar{x})^2} \text{ and } \hat{\beta}_0 = \bar{y} - \hat{\beta}_1 \bar{x}.$$

- Let's solve the first question together.
- Understanding the Data:

You are given the following data:

- **Weights (in pounds)**: 115, 117, 128, 125, 151, 162
- **Ages (in years)**: 18, 19, 20, 20, 21, 22

You need to find a **linear relationship** between weight (dependent variable y) and age (independent variable x).

Once you have the MLE formulas for estimating β_0 and β_1, you will:

1. Plug in the values of the weights (y) and ages (x) into the formulas.
2. Calculate β_0 (intercept) and β_1 (slope).

For example, the formulas for β_1 (slope) and β_0 (intercept) are typically derived as:

$$\beta_1 = \frac{\sum_{i=1}^{n}(x_i - \bar{x})(y_i - \bar{y})}{\sum_{i=1}^{n}(x_i - \bar{x})^2}$$

$$\beta_0 = \bar{y} - \beta_1 \bar{x}$$

Where:

- $\bar{x}$ is the mean of the ages,
- $\bar{y}$ is the mean of the weights.

- Interpreting the Results and Expected Increase

Once you have estimated β_0 and β_1, you can answer the question:

> **"If I increase my x (age) by 1 unit, what is the expected increase in y (weight)?"**

The answer is given by the **slope** β_1. In the context of this problem, β_1 represents the expected change in weight for each 1-year increase in age.

Thus, for each year increase in age, you expect the weight to increase by β_1 pounds. If β_1 is, for example, 2.5, it means that for each additional year of age, the weight is expected to increase by 2.5 pounds.

Practice Question IV:

9. **(10 points)** Roll a fair die (six faces).

   (a) **(5 points)** On average, how many rolls do I need to observe at least one even number and one odd number?

   (b) **(5 points)** On average, how many rolls do I need to observe all the numbers in $\{1, 3, 4\}$?

8. Roll a fair dice. On average, how many rolls do I need so that I observe all the numbers in $\{1, 2, 3, 4, 5, 6\}$. Hint: use the mean of Geometric distribution and the linearity of expectation.

2. **(25 points)** Roll three identical fair dices. On average, how many rolls do I need so that I observe all distinct combinations of outcomes from the three dices with numbers in $\{1, 2, 3, 4, 5, 6\}$. As a notice, the dices are labeled, for instance, if there is two 2 and one 3 among three dices, such senario indicates three different combinations: (2,2,3),(2,3,2),(3,2,2). Hint: You may use the mean of Geometric distribution and the linearity of expectation. You may also use the Harmonic Series Formula and you don't need to calculate the logarithm:

- These questions are from last year’s midterm, assignment, and quiz.
- They are all asking about how to use linearity of expectation to solve complex expectation problems.

- Let’s solve the second question together. The other two can follow the same idea.

Breaking Down the Process

We can analyze this in stages, counting how many additional flips are needed to observe a new number that hasn't been seen before.

1. **First outcome:** The first flip always introduces a new number. This takes exactly **1 flip**.
2. **Second outcome:** There are 5 remaining unseen numbers. The probability of getting a new number is $\frac{5}{6}$, so the expected number of flips required to get a new number is:

$$\frac{1}{\frac{5}{6}} = \frac{6}{5}$$

3. **Third outcome:** Now, 4 numbers remain unseen. The probability of getting a new number is $\frac{4}{6} = \frac{2}{3}$, so the expected number of flips required is:

$$\frac{1}{\frac{4}{6}} = \frac{6}{4} = \frac{3}{2}$$

4. **Fourth outcome:** There are 3 unseen numbers, with probability of getting a new number as $\frac{3}{6} = \frac{1}{2}$, so the expected number of flips is:

$$\frac{1}{\frac{3}{6}} = \frac{6}{3} = 2$$

5. **Fifth outcome:** There are 2 unseen numbers, with probability $\frac{2}{6} = \frac{1}{3}$, so the expected flips are:

$$\frac{1}{\frac{2}{6}} = \frac{6}{2} = 3$$

6. **Sixth outcome:** The final missing number appears with probability $\frac{1}{6}$, so the expected number of flips to get it is:

$$\frac{1}{\frac{1}{6}} = 6$$

Summing the Expected Values

The total expected flips required is:

$$E(T) = 1 + \frac{6}{5} + \frac{6}{4} + \frac{6}{3} + \frac{6}{2} + \frac{6}{1}$$
$$= 6\left(\frac{1}{1} + \frac{1}{2} + \frac{1}{3} + \frac{1}{4} + \frac{1}{5} + \frac{1}{6}\right)$$

Using the **harmonic series** approximation:

$$H_6 = 1 + \frac{1}{2} + \frac{1}{3} + \frac{1}{4} + \frac{1}{5} + \frac{1}{6} \approx 2.45$$
$$E(T) \approx 6 \times 2.45 = 14.7$$

Conclusion:

- The **linearity property** can be effectively utilized to compute the **expectation** of complex problems. By exploiting the linearity of expectation, we can decompose the expectation of a sum into the sum of individual expectations. This property simplifies the otherwise computationally intensive task of expectation calculation.

Midterm Review: Core Concept

Probability

- Sample Space, Events,…
- PMF, PDF, CDF
- Mean, Variance

Sample Space & Events

- **Random Experiment**: a repeatable procedure
- **Sample space**: set of all possible outcomes Ω.
- **Event**: a subset of the sample space.
- **Probability function**, P(ω): gives the probability for each outcome ω ∈ Ω
  - Probability is between 0 and 1
  - Total probability of all possible outcomes is 1.
  - If A=$\{\omega_1, \omega_2, \omega_3, \ldots\}$, P(A)=P($\omega_1$)+P($\omega_2$)+P($\omega_3$) $+ \cdots$

Sample Space

- Discrete or continuous: countable (listable) or not?
- A sample space is discrete if it consists of a finite or countable infinite set of outcomes.
- A sample space is continuous if it contains an interval (or a union of multiple intervals) of real numbers.

Sample space - example

- Consider the random experiment in which items are selected from a batch consisting of three items {a,b,c}

- Case 1: select two items **without replacement**

  Sample space $\{ab, ac, ba, bc, ca, cb\}$

- Case 2: select two items **with replacement**

  Sample space $\{aa, ab, ac, ba, bb, bc, ca, cb, cc\}$

Events

- Events are sets:
  - ✓ Can describe in words
  - ✓ Can describe in notation
- Experiment: toss a coin 2 times.
- Event -- You get 1 or more heads

  = {HH, HT, TH}

Set operations

- The commutative laws: $A \cup B = B \cup A$, $A \cap B = B \cap A$
- The associative laws: $(A \cup B) \cup C = A \cup (B \cup C)$, $(A \cap B) \cap C = A \cap (B \cap C)$
- The distributive laws: $A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$, $A \cup (B \cap C) = (A \cup B) \cap (A \cup C)$
- De Morgan's laws: $(A \cup B)' = A' \cap B'$, $(A \cap B)' = A' \cup B'$

Relation between two events

- Independence: Two events A and B are independent if and only if $P(A \cap B) \; = \; P(A) \cdot P(B)$

- Mutually Exclusive: Two events A and B are mutually exclusive if and only if $A \cap B \; = \emptyset$, which implies $P(A \cap B) = 0$

- A useful result: $P(A) = 1 - P(A')$

Random Variable

- X is called a random variable as it takes a numerical value that depends on the outcome of an experiment.

- Range of X: the set of possible values for X.

Probability Mass Function

- For a discrete random variable X with possible values $x_1, x_2, \dots, x_n$. A probability mass function $f(\cdot)$ is a function such that:

✓ $f(x_i) \geq 0$ for all $x_1, x_2, \dots, x_n$.
✓ $\sum_{i=1}^{n} f(x_i) = 1$
✓ $f(x_i) = \boxed{P(X = x_i)}$ for all $x_1, x_2, \dots, x_n$.

Probability that the random variable takes value x_i.

Probability Density Function

- For a continuous RV X, $P(X = x) = 0$ but $\{x\}$ is not an impossible event.

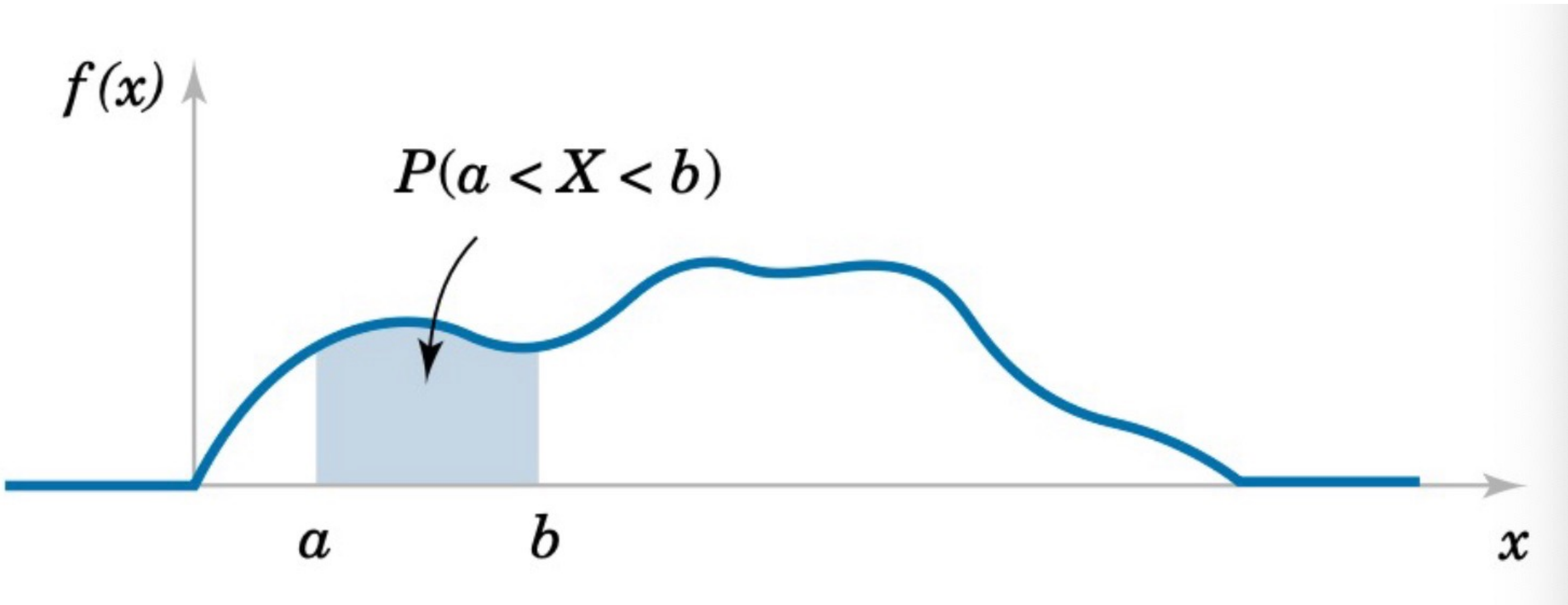
> - If $P(E) = 0$, then E is a zero-probability event.
> - If E is empty, then E is impossible.

- For a continuous random variable, we introduce a function f(ω), called the probability density function (pdf).
  - f(ω) > 0, if $\omega \in S$
  - f(ω) = 0, if $\omega \notin S$
  - $\int_{-\infty}^{\infty} f(x)dx = 1.$

Probability of $X \in [a, b]$

For continuous RV, the probability is defined based on intervals.

$$P(a \leq X \leq b) = \int_a^b f(x)dx$$



Cumulative Distribution Function

- The cumulative distribution function (cdf) gives the probability that the random variable X is less than or equal to x and is usually denoted by $F(x)$

- $F(x) = P(X \leq x)$

- $f(x) = F(x) - \lim_{y \uparrow x} F(y)$

Mean and Variance

- Mean

$$E[X] = \Sigma_x x\, P(X = x) = \Sigma_x x\, f(x)$$

- Variance

$$Var[X] = \Sigma_x (x - \mathrm{E}[X])^2\, f(x)$$

Linearity: $E[\sum_i C_i X_i] = \sum_i C_i E[X_i]$

C_i is a constant

No assumption on X_i

Expectation of a function of X:

$$\mathrm{E}[g(X)] = \sum_x g(x) P(X = x) = \sum_x g(x) f(x)$$

Expectation of a constant: $E[C] = C$

How do we use linearity to compute complex expectations?

Linearity: $\mathrm{E}[\sum_i X_i] = \sum_i \mathrm{E}[X_i]$

No assumption on X_i

Conditional Probability

- Given the realization of event A, the probability of event B may change

- (Conditional probability) If A and B are events with P(B)>0, then the conditional probability of A given B, denoted by P(A|B), is defined as $P(A|B) = \frac{P(A \cap B)}{P(B)}$

- (Independence) Two events A and B are called independent if and only if $P(A \cap B) = \mathrm{P}(A)P(B)$ $P(A|B) = \frac{P(A \cap B)}{P(B)} = \mathrm{P(A)}$

Discrete RV

- Bernoulli distribution
- Binomial distribution
- Geometric distribution

Bernoulli distribution

- Take value 1 with probability p and value 0 with probability 1 − p.
- $f(1) = p, f(0) = 1 - p$

Other Examples

- Toss a coin
- Win or lose in a game

What is $E[X_i]$ and $\mathrm{Var}(X_i)$?

- Mean=p
- Variance=p(1-p)

Binomial distribution

- With parameters n and p
- N independent experiments
- Each experiment: success (with probability p) or failure (with probability 1 − p).
- X: the number of success.

$$\Pr(X = k) = \binom{n}{k} p^k (1-p)^{n-k} \qquad \binom{n}{k} = \frac{n!}{k!(n-k)!}$$

- Mean=np
- Variance=np(1-p)

Geometric distribution

- Continuously draw a Bernoulli R.V.
- The X-th draw is the first success.
- X follows a geometric distribution.

$$\Pr(X = k) = (1 - p)^{k-1} p$$

- Mean=1/p
- Variance=(1-p)/p^2

Continuous RV

- Uniform Distribution
- Normal Distribution

Uniform Distribution

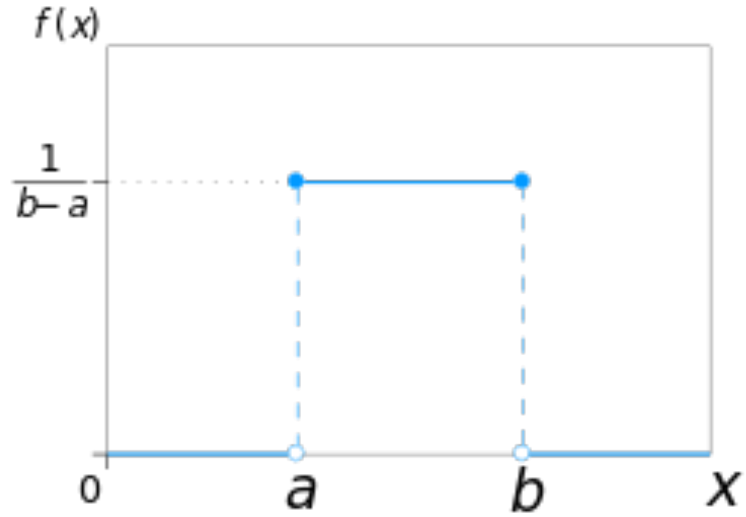
- With the "same probability", X takes a value within [a, b], where b>a.

- f(x) = c for $x \in [a, b]$ and f(x) = 0 for $x \notin [a, b]$
- As $\int_{-\infty}^{\infty} f(x)dx = c(b-a) = 1,$ we have

$$c = \frac{1}{b-a}$$

Uniform Distribution

- With the same probability, X takes a value within [a, b]
- $X \sim Uniform(a, b)$
- Mean= $(a + b)/2$
- Variance=$(b - a)^2/12$



Applications

How to approximate $\int_0^2 e^{x^2+\cos(x)}dx$?

Applications

- Given $X \sim Uniform(0,2)$
- What's the value of $E[2\ e^{X^2+\cos(X)}]$?

- f(x) = ½ for $x \in [0,2]$
- $E[2\ e^{X^2+\cos(X)}] = \int_0^2 2\ e^{x^2+\cos(x)} f(x) dx = \int_0^2\ e^{x^2+\cos(x)} dx$

Given $X \sim Uniform(0,2)$, $E[2\, e^{X^2+\cos(X)}] = \int_0^2 e^{x^2+\cos(x)} dx$

- Draw N samples of $X \sim Uniform(0,2)$: $X_1, X_2, X_3, \ldots, X_N$
- Calculate $\frac{\sum_i 2\, e^{X_i^2+\cos(X_i)}}{N}$

Why? Expectation can be approximated by long-run average

Law of Large Numbers

General Case

- How to calculate $\int_a^b h(x)dx$?

  - Draw N samples of $X \sim Uniform(a, b)$: $X_1, X_2, X_3, \ldots, X_N$
  - Calculate $\frac{\Sigma_i (\boldsymbol{b-a})\, h(X_i)}{N}$

    - $E[h(x)]$ only gives you the average "height" of $h(x)$
    - In order to get $\int_a^b h(x)dx$, which is the area, we need to multiply $E[h(x)]$ by $(b-a)$

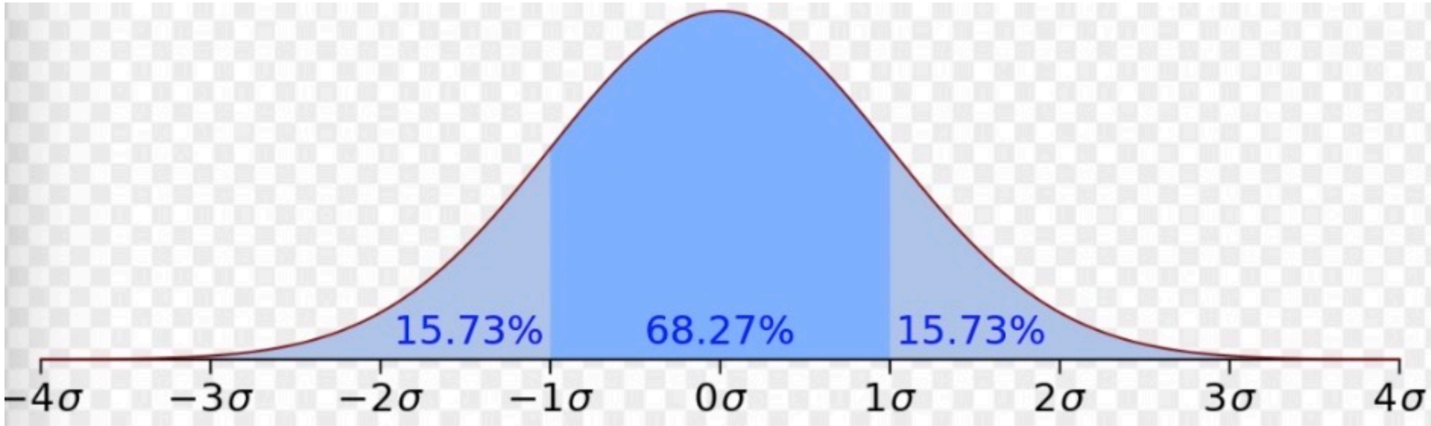
- Let $X \sim Uniform(a, b)$
- f(x) = 1/(b-a) for $x \in [a, b]$
- $E[(b-a)h(x)] = \int_a^b (b-a)h(x)f(x)dx = \int_a^b h(x)dx$

Normal Distribution

- X can be any real number
- Parameters: $\mu\ (mean) and\ \sigma^2\ (variance)$

$$f(x;\mu,\sigma)=\frac{1}{\sigma\sqrt{2\pi}}\exp\left(-\frac{(x-\mu)^2}{2\sigma^2}\right)。$$

- $X \sim Normal(\mu, \sigma^2)$



Statistics

- Maximum Likelihood Estimation
- Linear Regression

Maximum likelihood estimate (MLE)

- Target: estimate θ of a model

- Samples: $X_1,\ X_2\,, \ldots\,, X_n$
- Possible models: $\theta \in \Theta$ (Θ depends on the information you have)
- Model performance (probability): L(θ)= P ($X_1,\ X_2 \ldots,\ X_n | \theta$)

- Estimator: $\hat{\theta}$ such that L(θ) is maximized at θ= $\hat{\theta}$.

Likelihood function

- Given a model with an unknown parameter θ
- Given samples: $X_1,\ X_2 \ldots,\ X_n$

Continuous RV model:

- Likelihood: L(θ)= $\boxed{\prod_i}$ f($X_i\,|\,\theta$)
- Log-Likelihood: $l(\theta)$= $\sum_i \log(\mathrm{f}(X_i\,|\,\theta))$

Discrete RV model:

- Likelihood: L(θ)= $\boxed{\prod_i}$P ($X_i\,|\,\theta$)
- Log-Likelihood: $l(\theta)$= $\sum_i \log(\mathrm{P}(X_i\,|\,\theta))$

Why do we multiply the pdfs or pmfs here?

Because we assume those samples are independent observations

Example

- Suppose heights of people follow a normal distribution.
  - Given parameter μ, the model is $N(\mu, 1)$

- Samples: $X_1, \; X_2 \dots, \; X_n$

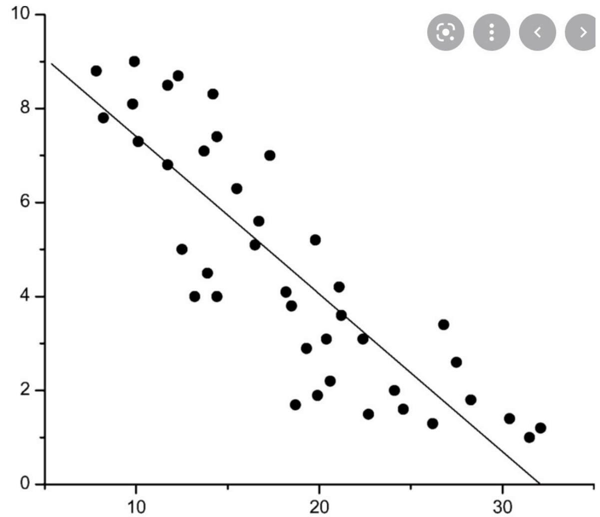
- $\mathrm{L}(\mu) = \left(\frac{1}{\sqrt{2\pi}}\right)^n e^{-\frac{1}{2}\sum_i (X_i - \mu)^2}$
- $l(\mu) = \mathrm{Log\ L}(\mu) = -\frac{1}{2}\sum_i (X_i - \mu)^2 - \frac{\mathrm{n}}{2}\log 2\pi$

- $l'(\mu) = \sum_i (X_i - \mu) = 0 \quad \Longrightarrow \quad \hat{\mu} = \bar{X}.$

Linear Regression

- Step 1: Propose a model ($Y \sim N(\beta_0 + \beta_1 X, \sigma^2)$)
- Step 2: Estimate β_0, β_1 (Maximum Likelihood Estimate)
- Step 3: Check assumptions (residual analysis)

Propose a model



- $Y \sim N(\beta_0 + \beta_1 X, \sigma^2)$

- Given the observation of X, Y follows a normal distribution with mean $\beta_0 + \beta_1 X$, and variance σ^2

- To simplify the analysis, we assume σ^2 is known

Estimate β_0, β_1

- Samples: (X_1, Y_1), … , (X_N, Y_N)
- For the model with β_0,β_1, σ^2 , the likelihood function is

$$\frac{1}{(\sqrt{2\pi})^n \sigma^n} \exp\left[-\frac{1}{2}\frac{\Sigma_i (Y_i - \beta_1 X_i - \beta_0)^2}{\sigma^2}\right]$$

- Given σ^2, to maximize the likelihood, we only need to minimize

$$\Sigma_i (Y_i - \beta_1 X_i - \beta_0)^2$$

- Taking derivative over β_0 and β_1, we have

$$\Sigma_i (Y_i - \beta_1 X_i - \beta_0) = 0$$

$$\Sigma_i (Y_i - \beta_1 X_i - \beta_0) X_i = 0$$

Estimate β_0, β_1

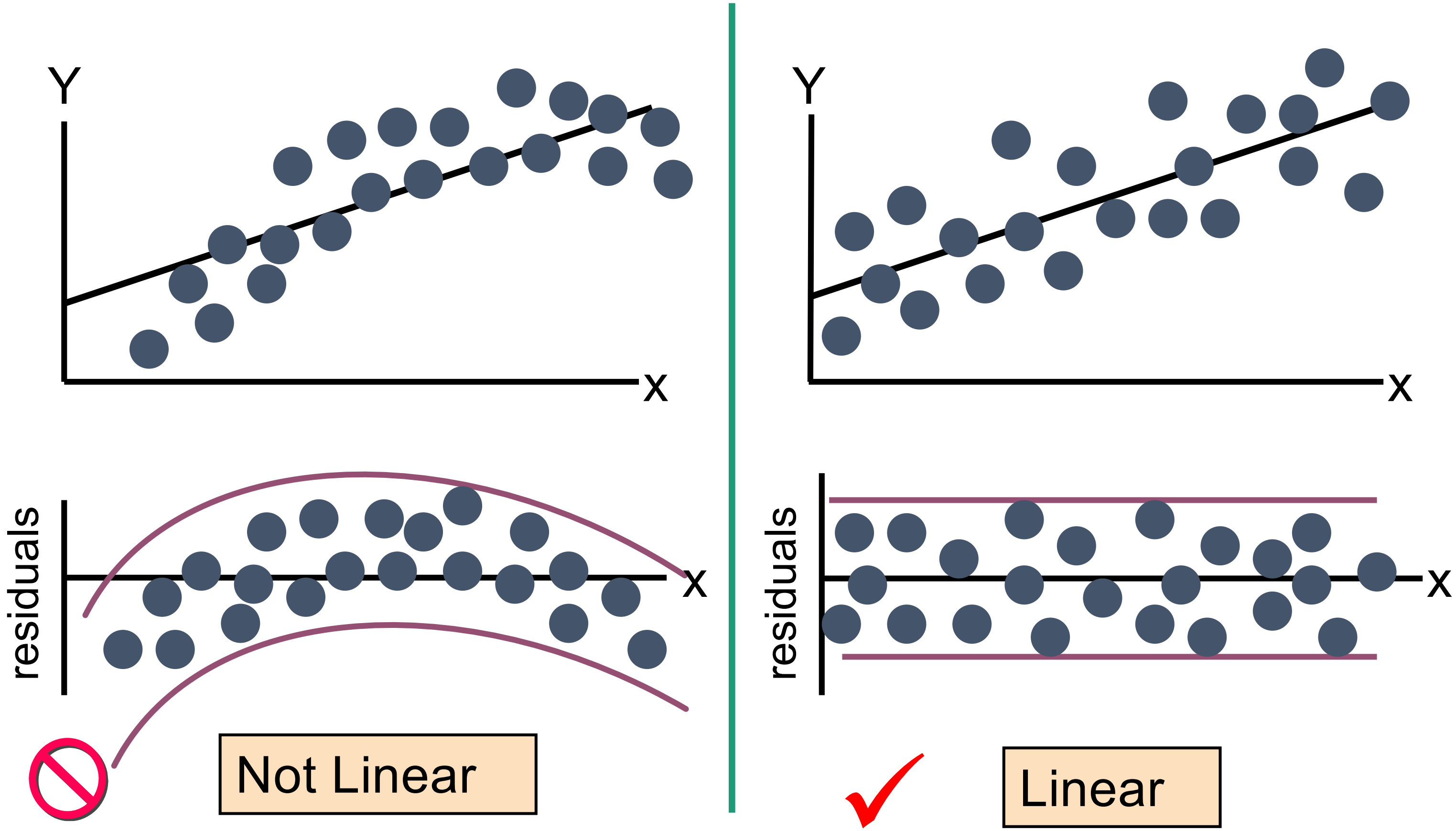
$$\Sigma_i(Y_i - \beta_1 X_i - \beta_0)X_i = 0 \text{ AND } \Sigma_i(Y_i - \beta_1 X_i - \beta_0) = 0$$

Eliminate β_0 first:

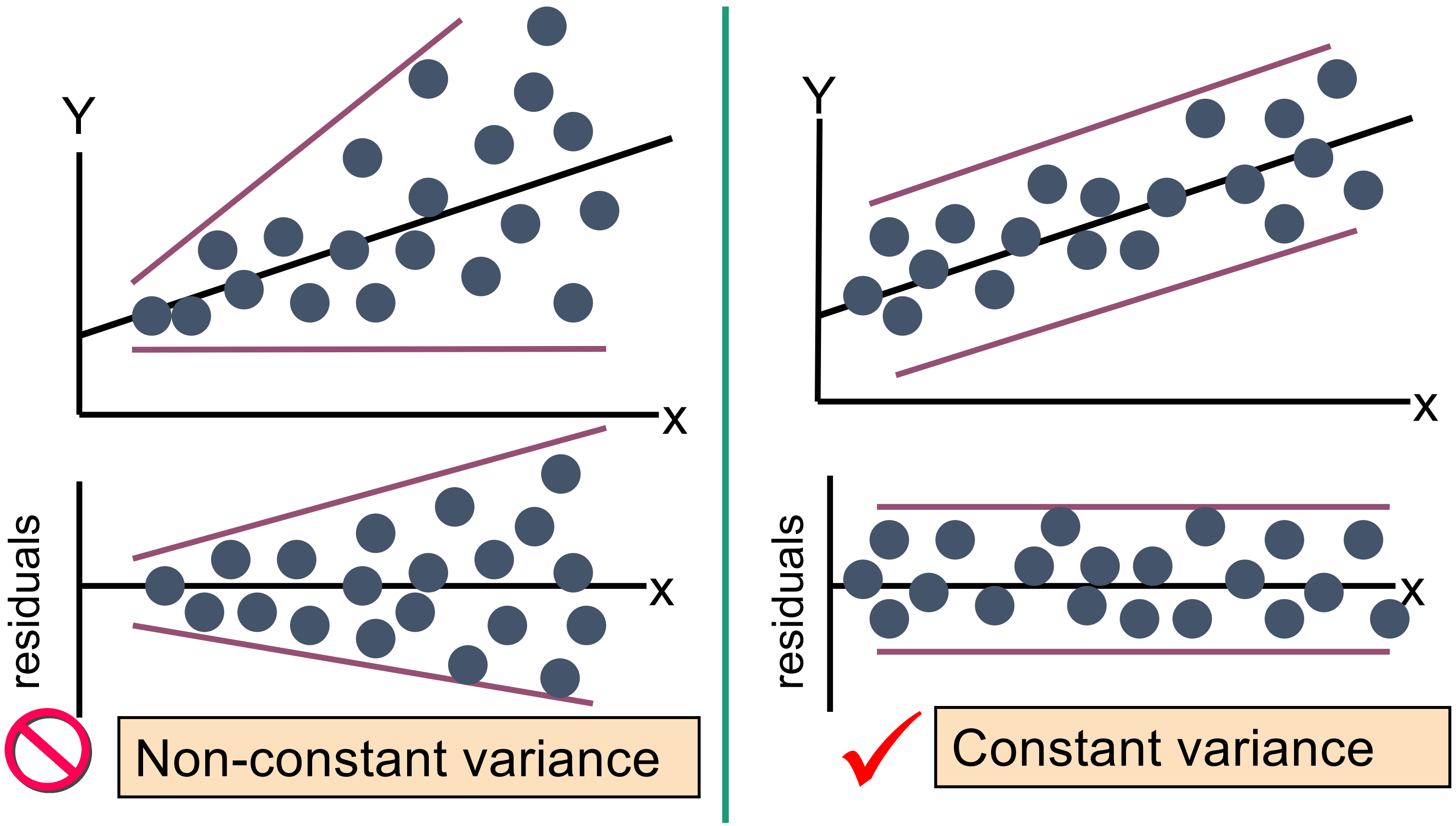
$$\Sigma_i(Y_i - \beta_1 X_i - \beta_0) = 0 \quad \text{->} \quad \beta_0 = \frac{1}{N}\Sigma_i(Y_i - \beta_1 X_i) = \bar{Y} - \beta_1 \bar{X}$$

MLE:
$$\left\{ \begin{array}{l} \widehat{\beta_1} = \dfrac{\Sigma_i(X_i - \bar{X})(Y_i - \bar{Y})}{\Sigma_i(X_i - \bar{X})^2} \\ \widehat{\beta_0} = \bar{Y} - \widehat{\beta_1}\bar{X} \end{array} \right.$$

Residual Analysis for Linearity



Residual Analysis for constant-variance



Interval Estimation

A random variable: X with variance $\boldsymbol{\sigma}^{\mathbf{2}}$

Data: $\mathbf{X_1}, \mathbf{X_2}, \cdots, \mathbf{X_n}$

Target: estimate the mean of the random variable.

Let $\bar{\mathbf{X}} = \frac{\mathbf{X_1}+\mathbf{X_2}+\cdots\mathbf{X_n}}{\mathbf{n}}$ be the sample mean, $\frac{\sqrt{\mathrm{n}}\,(\bar{\mathrm{X}}-\mu)}{\boldsymbol{\sigma}} \sim N(0,1)$ approximately

$$P(\bar{X} - \frac{b\,\sigma}{\sqrt{n}} \leq \mu \leq \bar{X} + \frac{a\,\sigma}{\sqrt{n}}) = P(-a \leq \frac{\sqrt{n}(\bar{X}-\mu)}{\boldsymbol{\sigma}} \leq b) = \Phi(b) + \Phi(a) - 1$$

Interval Estimation

- We usually let $a = b$ (the smallest interval)

- Define $\boldsymbol{z_{\alpha/2}}$ such that $\boldsymbol{P\left(-z_{\alpha/2} \leq Z \leq z_{\alpha/2}\right) = 1 - \alpha}$, where $\boldsymbol{Z \sim N(0, 1)}$
- Let b = $\boldsymbol{z_{\alpha/2}}$, a = $\boldsymbol{z_{\alpha/2}}$. Then $\Phi(b) - \Phi(-a)$ = 1- α.

- Then, [$\boldsymbol{\bar{X} - z_{\alpha/2}\frac{\sigma}{\sqrt{n}}, \bar{X} + z_{\alpha/2}\frac{\sigma}{\sqrt{n}}}$] is a 1- $\boldsymbol{\alpha}$ confidence interval.

Some observations

1-α Confidence Interval (CI):

$$\left[\bar{X} - z_{\alpha/2}\frac{\sigma}{\sqrt{n}}, \bar{X} + z_{\alpha/2}\frac{\sigma}{\sqrt{n}}\right]$$

The length of the confidence interval is affected by several factors

- As the sample size n **increases**, the length of CI **decreases**
- As the variance σ^2 **increases**, the length of CI **increases**
- As the confidence level increases (α **decreases**), the length of CI **increases**.

Point Estimation vs. Interval Estimation (Confidence Interval)

Point Estimation

- **Definition:** A single-value estimate of a population parameter (e.g., sample mean).
- **Example:** If the average test score in a sample is **75**, then the point estimate of the population mean is **75**.
- **Limitation:** Provides no information about uncertainty or variability in the estimate.

Interval Estimation (Confidence Interval)

- **Definition:** A range of values where the population parameter is likely to fall, with a specified confidence level (e.g., 95% CI).
- **Example:** A confidence interval of **(70, 80)** means we are **95% confident** that the true population mean lies between **70 and 80**.
- **Advantage:** Captures uncertainty and reflects variability in the data, making it more informative than point estimation.

Midterm Review

- I will hold online office hour every day from **March 11-March15** 9:00PM-10:00 PM (or by appointment)

Online zoom link:

https://cuhk-edu-cn.zoom.us/j/5533045922

Meeting ID: 553 304 5922

- The midterm will cover **Probability** and **Statistics**
- Please review our assignments, tutorials, and quiz, and core concep on our slides

Midterm Review

- Please send me a message by email (before Thursday) if you want me to explain any questions from our assignments, tutorials or quiz.
- My email: lishuang@cuhk.edu.cn

Practice Question I:

The random variable X has the following cumulative distribution function:

$$F(x) = \begin{cases} 0 & x < -2, \\ \beta & -2 \leq x < 0, \\ 10\beta^2 - 0.1 & 0 \leq x < 3, \\ 1 & x \geq 3. \end{cases}$$

where β is an unknown parameter. Suppose $E[X] = 1$.

Tasks

1. Determine the probability mass function (PMF) for X.
2. Compute $E[X^3 + \sin(X)]$.

Core concept behind this question

✓ CDF and PMF relationships
✓ Expectation computation for transformed variables
✓ Solving for unknown parameters using expectation constraints
✓ Properties of probability distributions

Step 1: Find the PMF

The PMF can be obtained from the CDF using:

$$p(x_i) = P(X = x_i) = F(x_i^+) - F(x_i^-).$$

By identifying the jump discontinuities in the CDF, we find the probabilities at each discrete value of X.

Step 1: Find the PMF

The PMF is obtained from the jumps in the cumulative distribution function (CDF), which occur at points where the function increases. These jumps correspond to the probability masses:

$$p(x) = P(X = x) = F(x^+) - F(x^-).$$

Identifying the jump points in $F(x)$:

- At $X = -2$:

$$p(-2) = F(-2^+) - F(-2^-) = \beta - 0 = \beta.$$

- At $X = 0$:

$$p(0) = F(0^+) - F(0^-) = (10\beta^2 - 0.1) - \beta = 10\beta^2 - \beta - 0.1.$$

- At $X = 3$:

$$p(3) = F(3^+) - F(3^-) = 1 - (10\beta^2 - 0.1) = 1 - 10\beta^2 + 0.1.$$

Thus, the PMF is:

$$P(X = -2) = \beta, \quad P(X = 0) = 10\beta^2 - \beta - 0.1, \quad P(X = 3) = 1 - 10\beta^2 + 0.1.$$

Step 2: Solve for β Using $E[X] = 1$

The expectation is given by:

$$E[X] = \sum_i x_i p(x_i) = 1.$$

We solve for β using this equation.

Step 2: Solve for β Using $E[X] = 1$

The expectation is given by:

$$E[X] = (-2)P(X = -2) + 0P(X = 0) + 3P(X = 3).$$

Substituting the PMF values:

$$E[X] = (-2\beta) + 0(10\beta^2 - \beta - 0.1) + 3(1 - 10\beta^2 + 0.1).$$

Simplifying:

$$E[X] = -2\beta + 3 - 30\beta^2 + 0.3.$$

Setting $E[X] = 1$:

$$-2\beta + 3 - 30\beta^2 + 0.3 = 1.$$

$$-30\beta^2 - 2\beta + 2.3 = 0.$$

Solving the quadratic equation:

$$30\beta^2 + 2\beta - 2.3 = 0.$$

Using the quadratic formula:

$$\beta = \frac{-2 \pm \sqrt{(2)^2 - 4(30)(-2.3)}}{2(30)}.$$

$$\beta = \frac{-2 \pm \sqrt{4 + 276}}{60}.$$

$$\beta = \frac{-2 \pm \sqrt{280}}{60}.$$

Approximating:

$$\sqrt{280} \approx 16.73.$$

$$\beta = \frac{-2 \pm 16.73}{60}.$$

$$\beta_1 = \frac{14.73}{60} \approx 0.2455, \quad \beta_2 = \frac{-18.73}{60} \approx -0.3122.$$

Since probabilities must be non-negative, we take:

$$\beta = 0.2455.$$

Step 3: Compute $E[X^3 + \sin(X)]$

Using the PMF found in Step 1, we compute:

$$E[X^3 + \sin(X)] = \sum_i \left(x_i^3 + \sin(x_i)\right) p(x_i).$$

Step 3: Compute $E[X^3 + \sin(X)]$

We compute:

$$E[X^3 + \sin(X)] = \sum_i \left(x_i^3 + \sin(x_i)\right) p(x_i).$$

Using the values $X = -2, 0, 3$:

$$E[X^3 + \sin(X)] = (-2)^3 P(X = -2) + \sin(-2) P(X = -2) + (0^3 + \sin 0) P(X = 0) + (3^3 + \sin 3) P(X = 3).$$

$$= (-8 + \sin(-2)) P(X = -2) + (0 + 0) P(X = 0) + (27 + \sin 3) P(X = 3).$$